

Ch 4: Quadratic Equations

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Check whether $x^2 - 5x + 6 = 0$ is a quadratic equation.

Q2. Check whether $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation.

Q3. Check whether $(x + 1)^2 = 2(x - 3)$ is a quadratic equation.

Q4. Check whether $x(x + 1) + 8 = (x + 2)(x - 2)$ is a quadratic equation. (PYQ)

Q5. Check whether $(2x - 1)(x - 3) = (x + 5)(x - 1)$ is a quadratic equation.

Q6. Find the roots of $x^2 - 3x - 10 = 0$ by factorisation.

Q7. Find the roots of $2x^2 + x - 6 = 0$ by factorisation.

Q8. Find the roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorisation. (PYQ 2019)

Q9. Find the roots of $100x^2 - 20x + 1 = 0$ by factorisation.

Q10. Find the roots of $4x^2 + 4\sqrt{3}x + 3 = 0$ by factorisation.

Q11. Find the roots of $x^2 - 45x + 324 = 0$ by factorisation.

Q12. Find the roots of $x^2 - x - 2 = 0$ by completing the square.

Q13. Find the roots of $5x^2 - 6x - 2 = 0$ using the quadratic formula.

Q14. Find the roots of $x^2 + 4x + 5 = 0$ using the quadratic formula and comment on the nature.

Q15. Find the discriminant of $2x^2 - 4x + 3 = 0$ and state the nature of roots.

Q16. Find the discriminant of $3x^2 - 2x + 1/3 = 0$ and state the nature of roots.

Q17. Find the discriminant of $x^2 - 4x + 4 = 0$ and state the nature of roots. (PYQ)

Q18. Find the value of k for which $2x^2 + kx + 3 = 0$ has two equal roots. (PYQ 2019)

Q19. Find the value of k for which $kx(x - 2) + 6 = 0$ has two equal roots. (PYQ 2019)

Q20. Find the value of k for which $x^2 - 2kx + 7k - 12 = 0$ has equal roots. (PYQ 2018)

Q21. Find the value of k for which $kx^2 - 2kx + 6 = 0$ has equal roots.

Q22. Find the value of k for which the equation $(k + 4)x^2 + (k + 1)x + 1 = 0$ has equal roots. (PYQ 2020)

Q23. Find the value of k for which $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ has equal roots.

Q24. If one root of $2x^2 + kx - 6 = 0$ is 2, find k and the other root. *(PYQ)*

Q25. If one root of $x^2 + kx + 4 = 0$ is -2 , find k .

Q26. What should be added to $x^2 - 5x + 4$ so that 3 becomes a zero of the resulting polynomial? *(PYQ 2024)*

Q27. Solve for x : $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$. *(PYQ 2020)*

Q28. Solve for x : $x + 1/x = 2.5$. *(PYQ)*

Q29. Solve for x : $(x - 1)/(x + 2) + (x - 3)/(x - 4) = 10/3$. *(PYQ)*

Q30. Solve for x : $4x^2 - 4ax + (a^2 - b^2) = 0$. *(PYQ 2019)*

Q31. The sum of two numbers is 27 and their product is 182. Find the numbers.

Q32. Find two consecutive positive integers whose product is 306. *(PYQ 2020)*

Q33. Find two consecutive odd positive integers whose sum of squares is 290. *(PYQ 2018)*

SECTION B - 3 MARKS

Q1. Find the roots of $4x^2 + 4\sqrt{3}x + 3 = 0$ by completing the square. (PYQ)

Q2. Find the roots of $5x^2 - 6x - 2 = 0$ by completing the square. (PYQ)

Q3. Find the roots of $2x^2 - 5x + 3 = 0$ by completing the square.

Q4. Find the roots of $3x^2 - 5x + 2 = 0$ by completing the square.

Q5. Find the roots of $x^2 + 4x - 5 = 0$ by completing the square and verify using factorisation. (PYQ)

Q6. Find roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ using the quadratic formula. (PYQ 2019)

Q7. Find roots of $3x^2 - 2\sqrt{6}x + 2 = 0$ using the quadratic formula and verify.

Q8. Solve: $9x^2 - 6b^2x - (a^4 - b^4) = 0$. (PYQ)

Q9. Solve: $x^2 - 2ax - (4b^2 - a^2) = 0$ using the quadratic formula.

Q10. Solve: $p^2x^2 + (p^2 - q^2)x - q^2 = 0$ using the quadratic formula. (PYQ)

Q11. Solve: $4x^2 - 4a^2x + (a^4 - b^4) = 0$ for x . (PYQ 2019)

Q12. Solve: $a(x^2 + 1) - x(a^2 + 1) = 0$.

Q13. Find the nature of roots: $2x^2 - 3x + 5 = 0$ and $3x^2 - 4\sqrt{3}x + 4 = 0$. (PYQ 2019)

Q14. Find the nature of roots of $2x^2 - 6x + 3 = 0$ and solve if real roots exist.

Q15. Find the nature of roots of $(x - 2a)(x - 2b) = 4ab$. (PYQ)

Q16. Find the value of k for which the quadratic equation $(k + 4)x^2 + (k + 1)x + 1 = 0$ has equal roots. (PYQ 2020)

Q17. Find k for which $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ has equal roots ($k \neq 12$).

Q18. Find k for which $kx^2 - 6x - 2 = 0$ has equal roots. Hence find the roots. (PYQ)

Q19. Find k for which $(k + 1)x^2 - 2(k - 1)x + 1 = 0$ has equal roots. (PYQ 2019)

Q20. Find k for which $x^2 + k(4x + k - 1) + 2 = 0$ has equal roots.

Q21. If $x = 2$ is a root of $x^2 - kx + 6 = 0$, find k . Also find the other root.

Q22. If $x = -2$ is a root of $3x^2 + 7x + p = 0$, find p . Also find the other root.

Q23. If the roots of $x^2 - 8x + k = 0$ are real and distinct, find k . (PYQ)

Q24. Prove that the roots of $x^2 + (a + b)x + c = 0$ are real if $(a - b)^2 \geq 4c - 4b^2$. (PYQ HOTS)

Q25. The product of two consecutive positive integers is 306. Find the integers. (PYQ 2020)

Q26. Find two consecutive odd positive integers whose sum of squares is 290. (PYQ 2018)

Q27. Find two numbers whose sum is 27 and product is 182. (PYQ 2019)

Q28. The sum of two numbers is 15. If the sum of their reciprocals is $\frac{3}{10}$, find the numbers. (PYQ 2020)

Q29. A two-digit number is such that the product of its digits is 35. When 18 is added to the number, the digits interchange. Find the number. (PYQ 2019)

Q30. A two-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange. Find the number. (PYQ 2018)

Q31. Five years later, the age of a person will be three times the age of his son. Five years ago, the person was seven times as old as his son. Find their present ages. (PYQ)

Q32. The sum of ages of a man and his son is 45. Five years ago, the product of their ages was 4 times the man's age at that time. Find their present ages. (PYQ 2020)

Q33. A motor boat whose speed is 18 km/hr in still water takes 1 hour more to go 24 km upstream than to return to the same point downstream. Find the speed of the stream. (PYQ 2019)

Q34. A train travels 360 km at uniform speed. If the speed had been 5 km/hr more, it would have taken 1 hour less. Find the speed of the train. (PYQ 2018)

Q35. A train travels 480 km at uniform speed. If the speed had been 8 km/hr less, it would have taken 3 more hours. Find the speed. (PYQ 2020)

Q36. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides. (PYQ 2018)

Q37. The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the sides. (PYQ 2019)

Q38. The area of a right-angled triangle is 600 cm^2 . If the base is 10 cm more than the altitude, find the base and altitude. (PYQ)

Q39. A rectangular park is to be designed whose breadth is 3 m less than its length. Its area is to be 4 square metres more than the area of a park that has already been made in the shape of an isosceles triangle with its base as the breadth of the rectangular park and altitude 12 m. Find its length and breadth. (PYQ 2019)

Q40. In a class test, the sum of Shefali's marks in Maths and English is 30. Had she got 2 more marks in Maths and 3 less in English, the product would have been 210. Find her marks in two subjects. (PYQ 2019)

Q41. Solve: $\frac{1}{(x+1)} + \frac{2}{(x+2)} = \frac{4}{(x+4)}$. (PYQ)

Q42. Solve: $\frac{1}{(x-3)} + \frac{2}{(x-2)} = \frac{8}{x}$. (PYQ)

Q43. Solve: $\frac{1}{a} + \frac{1}{b} + \frac{1}{x} = \frac{1}{(a + b + x)}$. (PYQ HOTS)

Q44. Solve: $\frac{(x - a)}{(x - b)} + \frac{(x - b)}{(x - a)} = \frac{a}{b} + \frac{b}{a}$. (PYQ HOTS)

SECTION C - 5 MARKS

Q1. A train travels 360 km at uniform speed. If the speed had been 5 km/hr more, it would have taken 1 hour less for the journey. Find the speed of the train. (PYQ 2018 — *most repeated*)

Q2. A motor boat whose speed is 18 km/hr in still water takes 1 hour more to go 24 km upstream than to return downstream. Find the speed of the stream. (PYQ 2019)

Q3. A train travels at a certain average speed for a distance of 63 km and then travels 72 km at an average speed of 6 km/hr more than the original speed. If it takes 3 hours in all, find the original average speed. (PYQ 2019)

Q4. An express train takes 1 hour less than a passenger train to travel 132 km between two stations. If the average speed of the express train is 11 km/hr more than that of the passenger train, find the speed of both. (PYQ 2020)

Q5. The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the sides of the field. (PYQ 2019)

Q6. Sum of areas of two squares is 468 m^2 . If the difference of their perimeters is 24 m, find the sides of the two squares. (PYQ 2018)

Q7. The sum of two numbers is 16 and sum of their cubes is 794. Find the numbers. (PYQ)

Q8. A park of rectangular shape has a perimeter of 80 m. A path 2.5 m wide running all around the inside of the park has an area of 225 m^2 . Find the dimensions of the park. (PYQ)

Q9. In a class test, the sum of Shefali's marks in Maths and English is 30. Had she got 2 more marks in Maths and 3 less in English, the product would have been 210. Find her marks. (PYQ 2019)

Q10. A two-digit number is such that the product of its digits is 35. When 18 is added to the number, the digits interchange. Also find a number 27 more than this number and verify if the new number is also a two-digit number with reversed digits. (PYQ 2019 extended)

Q11. Two water taps together can fill a tank in $9 \frac{3}{8}$ hours. The larger tap takes 10 hours less than the smaller one. Find the time each tap takes alone to fill the tank. (PYQ 2019)

Q12. Two pipes running together can fill a cistern in 3 and $1\frac{1}{13}$ minutes. If one pipe takes 3 minutes more than the other to fill it, find the time each pipe takes separately. (PYQ 2020)

Q13. A person on tour has Rs. 360 for his daily expenses. He decides to extend his tour programme by 4 days which means he will have to cut down his daily expenses by Rs. 3 per day. Find the original duration of the tour. (PYQ)

Q14. Rs. 6500 were divided equally among a certain number of persons. Had there been 15 more persons, each would have got Rs. 30 less. Find the original number of persons. (PYQ)

Q15. 300 apples are distributed equally among a certain number of students. Had there been 10 more students, each would have received one apple less. Find the number of students. (PYQ)

Q16. A pole has to be erected at a point on the boundary of a circular park of diameter 13 m in such a way that the differences of its distances from two diametrically opposite fixed gates A and B on the boundary is 7 m. Find the distances from the two gates.

Q17. The hypotenuse of a right triangle is 6 m more than twice the shortest side. If the third side is 2 m less than the hypotenuse, find the sides of the triangle. (PYQ)

Q18. A shopkeeper buys some books for Rs. 80. If he had bought 4 more books for the same amount, each book would have cost Rs. 1 less. Find the number of books originally bought. (PYQ 2020)

Q19. A plane left 30 minutes later than the scheduled time and in order to reach the destination 1500 km away on time, it had to increase its speed by 100 km/hr from the usual speed. Find its usual speed. *(PYQ 2019)*

Q20. A right triangle whose sides (other than hypotenuse) are 3 cm and 4 cm is made to revolve about its hypotenuse. Find the volume and surface area of the double cone so formed. Also apply the quadratic equation formed while finding the altitude of the triangle on the hypotenuse. *(HOTS)*

Q21. Find the roots of $4x^2 + 4bx - (a^2 - b^2) = 0$ by completing the square. Hence verify using the quadratic formula. *(PYQ 2019)*

Q22. Solve: $1/(x+4) - 1/(x-7) = 11/30$, $x \neq -4, 7$. *(PYQ 2020)*

Q23. Case Study: The speed of a boat in still water is x km/hr and the speed of the stream is y km/hr. It goes 30 km upstream and 44 km downstream in 10 hours. If it also goes 40 km upstream and 55 km downstream in 13 hours, form a quadratic equation and find x and y . Hence verify using the discriminant. *(PYQ case-based)*

Q24. A pole has to be erected at a point on the boundary of a circular park of diameter 13 m such that the difference of its distances from two diametrically opposite fixed gates A and B is 7 m. Find the distances from the two gates. *(PYQ)*

Q25. The sum of the squares of two consecutive odd numbers is 394. Find the numbers. Also verify that the numbers satisfy the

original condition. (PYQ)

Q26. A motor boat goes 10 km upstream and returns back to the starting point in 55 minutes. If the speed of the stream is 2 km/hr, find the speed of the motor boat in still water. Hence also find the total time taken if the speed of the stream increases to 3 km/hr. (PYQ)

Q27. Three consecutive positive integers are such that the sum of the square of the first and the product of the other two is 46. Find the integers. (PYQ)